

MP204 – Electricity & Magnetism

Problem Set 4

Due in the assignment box by 5pm, Friday 17 April 2015

1. A particle of charge q and mass m moves with velocity $\vec{v}$ in a magnetic field $\vec{B}$. Show that its total kinetic energy $K = \frac{1}{2}m|\vec{v}|^2$ is constant.
(Hint: compute dK/dt and express it in terms of the force $\vec{F}$ on the particle, then use the fact that $\vec{F} = q\vec{v} \times \vec{B}$.)
2. A thin wire of length L lies along the x -axis between $x = -L/2$ and $x = L/2$. A constant current I flows in the positive x -direction through this wire.

(a) Use the Biot-Savart law

$$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{r}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

to show that the magnetic field at the point $\vec{r} = a\hat{j}$ is

$$\vec{B} = \frac{\mu_0 I L}{2\pi a \sqrt{L^2 + 4a^2}} \hat{k}.$$

(b) Find the $L \rightarrow \infty$ limit of this field, and explain why the result of this limit has the form that it does.

(Feel free to look up any integrals that you feel you might need for this problem. Or even better, try to do them yourself!)

3. The *magnetic dipole moment* of a loop of current is defined to be

$$\vec{\mu} = \frac{I}{2} \oint_C \vec{r} \times d\vec{r},$$

where the line integral is computed around the loop in the direction of the current I .

- (a) Find the magnetic dipole moment of a circular loop with radius R .
- (b) Find the magnetic dipole moment of a square loop with side length a .
- (c) EXTRA CREDIT: Look at the ratios $\vec{\mu}/I$ for both (a) and (b); do they suggest a general form for the direction and magnitude of the dipole moment for a loop which lies in a plane?

(Hint: for (a) and (b), do the calculation assuming each loop lies in the xy -plane with the origin at its centre and the current flowing anticlockwise.)

4. The differential forms of Maxwell's equations are

$$\begin{aligned} (1) \vec{\nabla} \cdot \vec{E} &= \rho/\epsilon_0, & (2) \vec{\nabla} \cdot \vec{B} &= 0, \\ (3) \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} & (4) \vec{\nabla} \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}. \end{aligned}$$

Suppose we have an electromagnetic field given by

$$\vec{E} = E_0 (xz \sin \omega t - y^2 \cos \omega t) \hat{k}, \quad \vec{B} = \frac{E_0}{\omega} (2y \sin \omega t \hat{i} - z \cos \omega t \hat{j} + x \hat{k}),$$

where E_0 and ω are positive constants.

- (a) Confirm that these fields satisfy Maxwell's second and third equations.
- (b) Use Maxwell's first and fourth equations to find the charge density ρ and the current density $\vec{J}$.
- (c) Use your answers for (b) to confirm that the continuity equation

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$$

is satisfied.

VECTOR CALCULUS FORMULAE IN CARTESIAN COORDINATES

Cartesian coordinates (x, y, z) with constant unit direction vectors $\hat{i}, \hat{j}, \hat{k}$

- gradient of a function $f(x, y, z)$:

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

- divergence of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- curl of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{i} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{j} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{k}$$

- Laplacian of a function $f(x, y, z)$:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$